NADA EDDAHIR

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WORK EXPERIENCE

Coding Instructor - *GirlsCodeToo, Zürich, Switzerland* 2026 - PRESENT

- Teach hands-on coding workshops (Python, robotics, web development) to girls aged 8–18, as part of a Swiss non-profit working to close the gender gap in tech

Research Assistant - *ADAMMA Lab, ETH Zürich, Switzerland* 2026 - PRESENT

- Evaluate prediction stability of 5 LLMs against established uncertainty measures (e.g., semantic entropy) on 5 extractive-QA benchmarks to assess answer-invariant neighborhoods as a reliability signal

Research Assistant - *Distributed Computing Lab (DCL), EPFL, Lausanne, Switzerland* 2024 - 2025

- Designed and benchmarked 4 multi-agent coordination strategies for quantized Mistral-7B agents (confidence-weighted voting, multi-round debate with judge, split-knowledge debate, router-based specialization) against Mixtral-8x7B on AI2-ARC and MMLU, using CPU-only inference via llama.cpp
- Built a router-based system of 4 domain-specialized SLMs reaching 72.5% accuracy on mixed-domain questions vs. 47.5% for a generalist baseline with 15% lower inference time

Navigation Engineer - *EPFL Xplore Team (European Rover Challenge), Lausanne, Switzerland* 2024 - 2025

- Developed path-planning and obstacle-detection algorithms in ROS for the team's rover, which won 1st Place for Best Performance in Navigation at the European Rover Challenge 2025

Telecommunications Engineer - *EPFL Spacecraft Team (EST), Lausanne, Switzerland* 2023 - 2025

- Contributed to the development and performance analysis of the UHF communication module for the CHESS CubeSat mission
- Analyzed and optimized data transmission through link-budget analysis, covering telecommand uplinks and telemetry downlinks

Tutor - *Students 4 Students - EPFL, Lausanne, Switzerland* 2023 - 2024

- Tutored EPFL courses in Analysis, AICC I (Advanced Information Computation and Communication), Linear Algebra, and Programming (Java and C++)

EDUCATION

ETH Zürich, Zürich, Switzerland 2025 - PRESENT

- Master of Science in Computer Science

EPFL, Lausanne, Switzerland 2022 - 2025

- Bachelor of Science in Communication Systems

PROJECTS

Faithful LLMs for Evidence-Based QA (ETH AI Center - Projects in Machine Learning Research)

- Investigated whether DPO with attribution-based preference pairs improves citation faithfulness in evidence-based QA, across a unified 5-dataset benchmark (29,151 examples); compared SFT, DPO, and DPO-on-SFT training regimes

EndurAI: AI-Powered Sports Companion App

- Developed a Kotlin-based fitness application leveraging computer vision and machine learning to analyze workouts in real-time, providing form correction and performance metrics

ImgFS: Image-oriented File System

- Developed a custom image-oriented file system in C to efficiently store and manage large-scale images, inspired by Facebook's "Haystack" system
- Implemented functionalities like deduplication using SHA-256 hashing and created a CLI and web server for managing images over the network

Fashion-MNIST Classification with MLP, CNN, and Vision Transformer

- Built and compared deep learning models (MLP, CNN, and Vision Transformer) to classify images from the Fashion-MNIST dataset
- Achieved 92.3% accuracy with CNN by leveraging spatial feature extraction and advanced data preprocessing techniques

Javions: ADS-B Message Decoder and Visualization System

- Developed a Java application with a real-time decoding pipeline and GUI to track aircraft positions from ADS-B messages on an interactive map

ADDITIONAL INFORMATION

- **Technical Skills:**

Java, Scala, Python, C/C++, Kotlin, R, SQL; PyTorch, TensorFlow, scikit-learn, Hugging Face, llama.cpp; ROS, Git, Docker, Linux

- **Languages:**

French (C2), English (C2), Arabic (C2), Spanish (B1), German (Basics)

- **Extracurricular:**

• AGEPOLY (General Student's Association of EPFL) (2024 - 2026): Led sports event organization for the student association, including karate classes for women+
• Coaching EPFL (2023 - 2024): Mentored students on academic and personal development
• Moroccan Association for Learning Disabilities and Difficulties (AMTDA) (2019): Supported dyslexic primary-school students with tailored educational mentoring